

Week 31

UNIT-IV RECURRENCE RELATIONS

Generating Functions:

①

Consider a sequence of real numbers $a_0, a_1, a_2, \dots$. Let us denote this sequence by $\langle a_k \rangle$, $k=0, 1, 2, \dots$, this sequence, suppose there exists a function $f(x)$ whose expansion in a series of powers of x is as given below

$$\begin{aligned} G(x) \text{ or } f(x) &= a_0 + a_1 x + a_2 x^2 + \dots + a_{n-1} x^{n-1} + \dots \\ &= \sum_{k=0}^{\infty} a_k x^k \end{aligned}$$

Then $f(x)$ is called a generating function for the sequence $\{a_k\}$

Ex: 1

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots = \sum_{k=0}^{\infty} x^k$$

$f(x) = (1-x)^{-1}$ is a generating function for the sequence $1, 1, 1, 1, \dots$

Ex: 2

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots = \sum_{k=0}^{\infty} (-1)^k x^k$$

$f(x) = (1+x)^{-1}$ is a G.F for the sequence $1, -1, 1, -1, \dots$

Week 31

Generating functions of sequences (2)

① Find the sequences generated by the following functions.

$$(i) (3+x)^3 = 3^3 \left[1 + \frac{x}{3}\right]^3$$

$$= 27 \left[1 + \binom{3}{1} \left(\frac{x}{3}\right) + \binom{3}{2} \left(\frac{x}{3}\right)^2 + \binom{3}{3} \left(\frac{x}{3}\right)^3\right]$$

Using binomial expansion

$$[\because (1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k = \binom{n}{0} x^0 + \binom{n}{1} x^1 + \dots]$$

$$= 27 \left[1 + 3C_1 \frac{x}{3} + 3C_2 \frac{x^2}{9} + 3C_3 \frac{x^3}{27}\right]$$

$$= 27 \left[1 + x + \frac{3 \times 2}{2 \times 1} \frac{x^2}{9} + 1 \frac{x^3}{27}\right]$$

$$= 27 \left[1 + x + \frac{x^2}{3} + \frac{x^3}{27}\right]$$

$$= 27 + 27x + 9x^2 + x^3$$

$\therefore$ The sequence generated by $(3+x)^3$ is

$$27, 27, 9, 1, 0, 0, 0, \dots$$

— x —

$$(ii) 2x^2(1-x)^{-1} = G_1(x)$$

Sol:- Given

$$2x^2(1-x)^{-1} = 2x^2 [1 + x + x^2 + \dots]$$

$$= 2x^2 + 2x^3 + 2x^4 + \dots$$

$$= 0 + 0x + 2x^2 + 2x^3 + 2x^4 + 2x^5 + \dots \quad (3)$$

$\therefore$ The sequence generated by $2x^2(1-x)^{-1}$ is $0, 0, 2, 2, 2, \dots$

_____ X _____

(iii) $\frac{1}{1-x} + 2x^3$

Sol:- Given

$$\frac{1}{1-x} + 2x^3 = (1-x)^{-1} + 2x^3$$

$$= (1 + x + x^2 + x^3 + x^4 + \dots) + 2x^3$$

$$= 1 + x + x^2 + 3x^3 + x^4 + x^5 + \dots$$

$\therefore$ The sequence generated by $\frac{1}{(1-x)} + 2x^3$

is $1, 1, 1, 3, 1, 1, \dots$

_____ X _____

(iv) $3x^3 + e^{2x}$

Sol:- Given $[\because e^x = 1 + \frac{x}{1} + \frac{x^2}{2} + \dots]$

$$3x^3 + \left\{ 1 + \frac{(2x)}{1} + \frac{(2x)^2}{2} + \frac{(2x)^3}{6} + \dots \right\}$$

$$= 1 + \frac{2}{1}x + \frac{2^2}{2}x^2 + \left(3 + \frac{2^3}{6} \right)x^3 + \frac{2^4}{24}x^4 + \dots$$

$\therefore$ The sequence generated by $3x^3 + e^{2x}$ is

$1, \frac{2}{1}, \frac{2^2}{2}, \left(3 + \frac{2^3}{6} \right), \frac{2^4}{24}, \dots$

_____ X _____

Find the generating functions for the following sequences

(i) 1, 2, 3, 4, ...

Sol:- By binomial expansion

$$(1-x)^{-2} = 1^0 + 2x^1 + 3x^2 + 4x^3 + 5x^4 + \dots$$

$\therefore f(x) = (1-x)^{-2}$ is a generating function for the sequence 1, 2, 3, 4, ...

(ii) 1, -2, 3, -4, ...

Sol:- By binomial expansion

$$(1+x)^{-2} = 1^0 - 2x^1 + 3x^2 - 4x^3 + 5x^4 - 6x^5 + \dots$$

$\therefore f(x) = (1+x)^{-2}$ is a G.F for the sequence 1, -2, 3, -4, ...

(iii) 0, 1, 2, 3, ...

Sol:- Given. 0, 1, 2, 3, ...

$$0 + 1x + 2x^2 + 3x^3 + \dots = x[1 + 2x + 3x^2 + \dots] = x[1-x]^{-2}$$

$f(x) = x(1-x)^{-2}$ is a G.F for the sequence 0, 1, 2, 3, ...

(5)

$$(iv) 0, 1, -2, 3, -4, \dots$$

$$0 + 1x - 2x^2 + 3x^3 - 4x^4 + \dots$$

$$= x[1 - 2x + 3x^2 - 4x^3 + \dots]$$

$$\therefore f(x) = x[1 + x]^{-2}$$

_____ X _____.

$$(v) 1^2, 2^2, 3^2, \dots$$

using

$$0 + 1x + 2x^2 + 3x^3 + \dots = x(1-x)^{-2}$$

D.w.r. to x we get

$$1 + 2(2x) + 3(3x^2) + \dots = \frac{d}{dx} \left[\frac{x}{(1-x)^2} \right]$$

$$= \frac{(1-x)^2(1) - x(2(1-x)(-1))}{(1-x)^4}$$

$$= \frac{(1-x)^2 + 2x(1-x)}{(1-x)^4}$$

$$= \frac{(1-x)[1-x+2x]}{(1-x)^4}$$

$$1^2 + 2^2x + 3^2x^2 + \dots = \frac{1+x}{(1-x)^3}$$

$\therefore f(x) = \frac{1+x}{(1-x)^3}$ is a G.F for the
sequence $1^2, 2^2, 3^2, \dots$

_____ X _____.

(6)

$$(vi) 0^2, 1^2, 2^2, 3^2, \dots$$

Sol:- Using

$$0^2 + 1^2x + 2^2x^2 + 3^2x^3 + \dots = x[1^2 + 2^2x + 3^2x^2 + \dots]$$

$$= x \left[\frac{1+x}{(1-x)^3} \right] \text{ by using the result (v) above}$$

$$f(x) = \frac{x(1+x)}{(1-x)^3}$$

$\therefore f(x) =$ is a G.F for the sequence

$$0^2, 1^2, 2^2, 3^2, \dots$$

_____ X _____

Find the G.F for the following sequences

$$(i) 1, 1, 0, 1, 1, 1, \dots$$

Sol:- Here

$$a_0=1, a_1=1, a_2=0, a_3=1, a_4=1, a_5=1, \dots$$

A G.F for this sequence is given by

$$\begin{aligned} \text{G.F } f(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + \dots \\ &= 1 + x + 0x^2 + 1x^3 + 1x^4 + 1x^5 + \dots \\ &= 1 + x + x^3 + x^4 + x^5 + \dots \\ &= 1 + x + x^2 - x^2 + x^3 + x^4 + x^5 + \dots \\ &= 1 + x + x^2 + x^3 + x^4 + \dots - x^2 \\ &= (1-x)^{-1} - x^2 \\ &= \frac{(1-x)^{-1} - x^2}{x} \end{aligned}$$

_____ X _____

(ii) $0, 2, 6, 12, 20, 30, 42, \dots$

Soln - $a_0 = 0, = 0 + 0^2$

$a_1 = 2, = 1 + 1^2$

$a_2 = 6, = 2 + 2^2$

$a_3 = 12 = 3 + 3^2$

$a_4 = 20, = 4 + 4^2 \dots$

$a_5 = 30, = 5 + 5^2 \dots$

$\therefore a_r = r + r^2, r = 0, 1, 2, \dots$

$G(x) = f(x) = (0 + 1x + 2x^2 + 3x^3 + 4x^4 + \dots) + (0^2 + 1^2x + 2^2x^2 + 3^2x^3 + \dots)$

$= x(1 + 2x + 3x^2 + 4x^3 + \dots) + \frac{x(1+x)}{(1-x)^3}$

$= x(1-x)^{-2} + \frac{x(1+x)}{(1-x)^3}$

$= \frac{x}{(1-x)^2} + \frac{x(1+x)}{(1-x)^3}$

$= \frac{x(1-x) + x + x^2}{(1-x)^3}$

$\therefore f(x) = \frac{2x}{(1-x)^3}$

————— X —————

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Result:-

$$1. \text{ The G.F } f(x) = \sum_{r=0}^{\infty} a_r x^r \quad (8)$$

$$= 1 + x + x^2 + \dots + x^n = \frac{1-x^{n+1}}{1-x} \quad |x| < 1.$$

2. If n is a positive integer then

$$\binom{-n}{r} = (-1)^r \binom{n+r-1}{r}$$

$$= (-1)^r \binom{n+r-1}{n-1}$$

$$(ii) \binom{2n}{n} = \sum_{r=0}^n \binom{n}{r}^2$$

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$$(iii) (1+x)^{2n} = (1+x)^n \cdot (1+x)^n$$

$$= \sum_{r=0}^n \binom{n}{r} x^r \cdot \sum_{s=0}^n \binom{n}{s} x^s$$

(3) If n is a positive integer then

$$(1-x)^{-n} = \sum_{r=0}^{\infty} \binom{n+r-1}{r} x^r$$

————— $\times$ —————

CALCULATING COEFFICIENTS OF G.F

(9)

Some standard Results.

$$1. (1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$$

$$2. (1-x)^n = \sum_{r=0}^n \binom{n}{r} (-x)^r$$

$$3. (1-x)^{-n} = \sum_{r=0}^{\infty} \binom{n+r-1}{r} x^r \quad \text{if } n \text{ is a +ve integer}$$

$$4. 1+x+x^2+\dots+x^n = \frac{1-x^{n+1}}{1-x}, \quad x \neq 1 \text{ (G.S.)}$$

—————*—————

① Determine the coefficient of x^{12} in $\frac{x^3(1-2x)^{10}}{G.F.}$

Sol:- Given $n=10$

$$(a) x^3(1-2x)^{10} = x^3 \sum_{r=0}^{10} \binom{10}{r} (-2x)^r \quad \text{Here } n=10$$

$$= x^3 \sum_{r=0}^{10} \binom{10}{r} (-2)^r x^r \quad \left[\because \sum_{r=0}^n \binom{n}{r} (-2)^r = (1-2)^n \right]$$

$$= \sum_{r=0}^{10} (-2)^r \binom{10}{r} x^{r+3} = (-2)^0 \binom{10}{0} x^{0+3} + \dots + (-2)^9 \binom{10}{9} x^{9+3}$$

Put $r=9$.

The coefficient of x^{12} is

$$\begin{aligned} x^3(1-2x)^{10} &= (-2)^9 \binom{10}{9} = (-2)^9 10C_9 = -10(2^9) \\ &= -5120 // - \end{aligned}$$

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(b) x^0 in $(3x^2 - (\frac{2}{x}))^{15}$

(10)

Sol:-

$$(3x^2 - (\frac{2}{x}))^{15} = (3x^2)^{15} \left[1 - \left(\frac{2}{3x^3}\right)\right]^{15}$$

$$= 3^{15} x^{30} \sum_{r=0}^{15} \binom{15}{r} \left(-\frac{2}{3x^3}\right)^r$$

$$[\because (1-x)^n = \sum_{r=0}^n \binom{n}{r} (-x)^r]$$

$$= 3^{15} x^{30} \sum_{r=0}^{15} \left(-\frac{2}{3}\right)^r \binom{15}{r} (x^{-3r})$$

$$= 3^{15} \sum_{r=0}^{15} \left(-\frac{2}{3}\right)^r \binom{15}{r} x^{30-3r}$$

$$= 3^{15} \left[\left(-\frac{2}{3}\right)^0 \binom{15}{0} x^{30-0} + \left(-\frac{2}{3}\right)^1 \binom{15}{1} x^{30-3} + \dots + \left(-\frac{2}{3}\right)^{10} \binom{15}{10} x^{30-30} + \dots \right]$$

Put $r=10$ The coefficient of x^0 is

$$\left(3x^2 - \frac{2}{x}\right)^{15} = 3^{15} \left(-\frac{2}{3}\right)^{10} \binom{15}{10}$$

$$= 14348909 \times \frac{1024}{59049} \times 3003$$

$$= 74,72,42,496.$$

(c) x^5 in $(1-2x)^{-7}$

Sol:- Now

$$(1-2x)^{-7} = \sum_{r=0}^{\infty} \binom{7+r-1}{r} (2x)^r$$

$$[\because (1-x)^{-n} = \sum_{r=0}^{\infty} \binom{n+r-1}{r} x^r \text{ if } n \text{ is a positive integer.}]$$

$$\Rightarrow \sum_{r=0}^{\infty} \binom{6+r}{r} (2x)^r = \binom{6+0}{0} (2x)^0 + \dots + \binom{6+5}{5} (2x)^5 + \dots$$

put $r=5$

the coefficient of x^5 is

$$= \binom{6+5}{5} (2)^5$$

$$= \binom{11}{5} 2^5 = 11C_5 2^5$$

$$= 32 \frac{11!}{16!5!} = 14,784.$$

————— $\times$ —————

(d) x^{10} in $(x^3 - 5x) / (1-x)^3$

Sol:

$$\frac{x^3 - 5x}{(1-x)^3} = (x^3 - 5x)(1-x)^{-3}$$

$$= (x^3 - 5x) \sum_{r=0}^{\infty} \binom{3+r-1}{r} x^r$$

$$= (x^3 - 5x) \sum_{r=0}^{\infty} \binom{2+r}{r} x^r$$

$$= \sum_{r=0}^{\infty} \binom{2+r}{r} x^{r+3} - 5 \sum_{r=0}^{\infty} \binom{2+r}{r} x^{r+1}$$

$\Rightarrow \binom{2}{0} x^{0+3} + \binom{2+1}{1} x^{1+3} + \dots - 5 \left[\binom{2}{0} x^{0+1} + \dots + \binom{2+9}{9} x^{9+1} + \dots \right]$

Put $r=7$, Put $r=9$

(12)

The coefficient of x^{10} in the R.H.S is

$$C_{10} = \binom{9}{7} - 5 \binom{11}{9} = \frac{19}{12 \cdot 17} - 5 \left[\frac{11}{12 \cdot 17} \right]$$

$$= -239.$$

_____ $\times$ _____.

$$(e) \frac{(1+x)^4}{(1-x)^4} = (1+x)^4 (1-x)^{-4} = \sum_{r=0}^{\infty} \binom{4}{r} x^r (1-x)^{-4}$$

$$= (1+4x+6x^2+4x^3+x^4) \times (1-x)^{-4}$$

$$= (1+4x+6x^2+4x^3+x^4) \sum_{r=0}^{\infty} \binom{3+r}{r} x^r$$

16 Sunday $= (1+4x+6x^2+4x^3+x^4) \left[\binom{3}{0}x^0 + \binom{3+1}{1}x^1 + \binom{3+2}{2}x^2 + \binom{3+3}{3}x^3 + \binom{3+4}{4}x^4 + \dots \right]$

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equating the coefficient of x^{15}

but $r=15$

$$f(x) = \binom{18}{15} + 4 \binom{17}{14} + 6 \times \binom{16}{13} + 4 \times \binom{15}{12} + \binom{14}{11}$$

$$= 9080.$$

_____ $\times$ _____.

$$(f) \text{ we have } x^{20} \text{ in } (x^2+x^3+x^4+x^5+x^6)^5$$

now

$$(x^2+x^3+x^4+x^5+x^6)^5 = (x^2)^5 [1+x+x^2+x^3+x^4]^5$$

$$= x^{10} \left[\frac{1-x^5}{1-x} \right]^5 = x^{10} (1-x^5)^5 \cdot (1-x)^{-5}$$

$$\left[\because \text{G.S } 1+x+x^2+\dots+x^n = \frac{1-x^{n+1}}{1-x} \right]$$

$\frac{(1-x)^{-5}}{1-x}$

$$= x^{10} \sum_{r=0}^5 \binom{5}{r} (-x^5)^r \times \sum_{s=0}^{\infty} \binom{5+s-1}{s} x^s$$

$$\left[\because (1-x)^{-n} = \sum_{r=0}^{\infty} \binom{n+r-1}{r} x^r \right]$$

equating the coefficient of x^{20}

$$= x^{10} \sum_{r=0}^5 \binom{5}{r} (-x^5)^r \times \sum_{s=0}^{\infty} \binom{4+s}{s} x^s$$

$$= x^{10} \left[\binom{5}{0} x^0 + \binom{5}{1} (-x^5)^1 + \binom{5}{2} (-x^5)^2 + \dots \right. \\ \left. \binom{5}{3} (-x^5)^3 + \binom{5}{4} (-x^5)^4 + \binom{5}{5} (-x^5)^5 \right]$$

$$\sum_{s=0}^{\infty} \binom{4+s}{s} x^s \quad \begin{matrix} s=10 \\ s=5 \\ s=0 \end{matrix}$$

put $s=10$

$$= \binom{5}{0} \binom{4+10}{10} - \binom{5}{1} \binom{4+5}{5} + \binom{5}{2} \binom{4+0}{0}$$

$$= \binom{5}{0} \binom{14}{10} - \binom{5}{1} \binom{9}{5} + \binom{5}{2} \binom{4}{0}$$

$$= 1001 - 630 + 10$$

$$= 381$$

H.W Find the coefficient of x^{27} in the following

he (i) $(x^4 + x^5 + x^6 + \dots)^5$ (ii) $(x^4 + 2x^5 + 3x^6 + \dots)^5$

Week 31

① Find the coefficient of x^{18} in the following ⁽¹⁴⁾

Products:

$$(i) (x + x^2 + x^3 + x^4 + x^5)(x^2 + x^3 + x^4 + \dots)^5$$

$$(ii) (x + x^3 + x^5 + x^7 + x^9)(x^3 + 2x^4 + 3x^5 + \dots)^3$$

Sol:- (i) we have

$$\begin{aligned} & (x + x^2 + x^3 + x^4 + x^5)(x^2 + x^3 + x^4 + \dots)^5 \\ &= x^{11} (1 + x + x^2 + x^3 + x^4)(1 + x + x^2 + \dots)^5 \\ &= x^{11} (1 + x + x^2 + x^3 + x^4) \{ (1 - x)^{-1} \}^5 \\ &= x^{11} (1 + x + x^2 + x^3 + x^4) (1 - x)^{-5} \end{aligned}$$

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$$\begin{aligned} &= x^{11} (1 + x + x^2 + x^3 + x^4) \sum_{r=0}^{\infty} \binom{5+r-1}{r} x^r \\ &= x^{11} (1 + x + x^2 + x^3 + x^4) \sum_{r=0}^{\infty} \binom{4+r}{r} x^r \\ &= (x^{11} + x^{12} + x^{13} + x^{14} + x^{15}) \left[\binom{4+0}{0} x^0 + \binom{4+1}{1} x^1 + \binom{4+2}{2} x^2 + \binom{4+3}{3} x^3 + \binom{4+4}{4} x^4 + \dots \right] \end{aligned}$$

The coefficient of x^{18} in this is

$$= \binom{11}{7} + \binom{10}{6} + \binom{9}{5} + \binom{8}{4} + \binom{7}{3}$$

(ii) we have

$$\begin{aligned} & (x + x^3 + x^5 + x^7 + x^9)(x^3 + 2x^4 + 3x^5 + \dots)^3 \\ &= x(1 + x^2 + x^4 + x^6 + x^8) x^9 (1 + 2x + 3x^2 + \dots)^3 \end{aligned}$$

(15)

$$= x^{10} (1 + x^2 + x^4 + x^6 + x^8) \{ (1 - x)^{-2} \}^3$$

$$= x^{10} (1 + x^2 + x^4 + x^6 + x^8) (1 - x)^{-6}$$

$$= x^{10} (1 + x^2 + x^4 + x^6 + x^8) \sum_{r=0}^{\infty} \binom{6+r-1}{r} x^r$$

$$= x^{10} (1 + x^2 + x^4 + x^6 + x^8) \sum_{r=0}^{\infty} \binom{5+r}{r} x^r$$

The coefficient of x^{18} in this is

$$= (x^{10} + x^{12} + x^{14} + x^{16} + x^{18}) \left[\binom{5+0}{0} x^0 + \dots + \binom{5+2}{2} x^2 + \binom{5+4}{4} x^4 + \binom{5+6}{6} x^6 + \binom{5+8}{8} x^8 \right]$$

$$= \binom{13}{8} + \binom{11}{6} + \binom{9}{4} + \binom{7}{2} + \binom{5}{0}$$

$$= \binom{13}{8} + \binom{11}{6} + \binom{9}{4} + \binom{7}{2} + \binom{5}{0}$$

RECURRENCE RELATIONS.

Def:

A recurrence relation for the sequence

$\{a_n\}$ is an equation that expresses an terms of one or more of the previous terms of the sequence, namely $a_0, a_1, a_2, a_3, \dots, a_{n-1}$ for all integers n with $n \geq n_0$ where " n_0 " is a non-negative integer.

A sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

— α —

(16)

FIRST-order Recurrence Relations

We consider for solution recurrence relations of the form

$$a_n = c a_{n-1} + f(n), \quad n \geq 1$$

where c is a known constant and $f(n)$ is a known function. Such a relation is called a linear recurrence relation of first-order constant coefficient. If $f(n) = 0$ the relation is called homogeneous, otherwise it is called non-homogeneous (or inhomogeneous).

Note: If $f(n) = 0$ that is if the recurrence relation is homogeneous the solution becomes

$$a_n = c^n a_0, \quad \text{for } n \geq 1$$

① Solve the recurrence relation $a_{n+1} = 4a_n$ for $n \geq 0$ given that $a_0 = 3$.

Sol:- The given equation

$$a_{n+1} = 4a_n$$

This equation is homogeneous because $f(n) = 0$.

Its general eqn. is

(17)

$$a_n = 4^n a_0, \quad n \geq 1$$

$$\text{Put } a_0 = 3$$

$$a_n = 3(4^n), \quad \text{for } n \geq 1.$$

————— X —————

(2) Solve the recurrence relation $a_n = 7a_{n-1}$, $n \geq 1$ given that $a_2 = 98$.

Sol:- Given $a_n = 7a_{n-1}$, $n \geq 1$

Replace
Put $n = n+1$
↓
 n by $n+1$
This equation can be written as
 $a_{n+1} = 7a_n$, $n \geq 0$
w.k.t

$$a_n = c^n a_0, \quad n \geq 1$$

$$a_n = 7^n a_0 \rightarrow (1) \quad [\text{Here } c=7]$$

$$\text{Given } a_2 = 98 \rightarrow (2)$$

$$\text{from (1) } n=2$$

$$a_2 = 7^2 a_0$$

$$a_2 = 49 a_0 \rightarrow (3)$$

$$\text{from (2) \& (3)}$$

$$a_0 49 = 98$$

$$a_0 = 2$$

$$\text{Subst. in (1), } a_n = 7^n (2), \quad n \geq 1.$$

③ Solve the recurrence relation

$a_n - 3a_{n-1} = 5 \times 2^n$ for $n \geq 1$ given that $a_0 = 2$.

Sol: Given $a_n - 3a_{n-1} = 5 \times 2^n \rightarrow (1)$

① can be written as

$$\text{Put } n=n+1, \quad a_{n+1} - 3a_n = 5 \times 2^{n+1}$$

$$\begin{aligned} \text{[by changing } n \text{ to } n+1] \quad a_{n+1} &= 3a_n + 5 \times 2^{n+1} \\ &= 3a_n + f(n+1) \text{ where } f(n) \\ &= 5(2)^n \end{aligned}$$

9 Sunday which is the non-homogeneous equation.

W.K.T

The general solution of non-homogeneous relation is

$$a_n = c^n a_0 + \sum_{k=1}^n c^{n-k} f(k), \quad n \geq 1$$

Here $c=3$.

$$a_n = 3^n a_0 + \sum_{k=1}^n 3^{n-k} f(k)$$

Given $a_0 = 2$

$$a_n = 2(3)^n + 3^{n-1} f(1) + 3^{n-2} f(2) + \dots + 3^0 f(n)$$

Substituting for $f(n)$, $n=1,2,3,\dots$ (19)

This becomes $f(n) = 5 \times 3^n$

$$\begin{aligned} a_n &= 2 \times 3^n + 3^{n-1} (5 \times 3^1) + 3^{n-2} (5 \times 3^2) \\ &\quad + 3^{n-3} (5 \times 3^3) + \dots + 3^0 (5 \times 3^n) \\ &= 2(3)^n + 5 \{ 3 \times 3^{n-1} + 3^2 \times 3^{n-2} + \dots + 3^n \} \\ &= 2(3)^n + 5 \{ 3^n + 3^n + \dots + 3^n \} \\ &= 2(3)^n + 5(n \times 3^n) \\ &= (2+5n)(3)^n \end{aligned}$$

which is the required soln.

④ Find a recurrence relation and the initial condition for the sequence 2, 10, 50, 250, ... Hence find the general term of the sequence.

Sol: Given $a_0 = 2, a_1 = 10, a_2 = 50, a_3 = 250, \dots$

$$\text{Now } a_1 = 5a_0 = 5 \times 2$$

$$a_2 = 5a_1 = 5 \times 10$$

$$a_3 = 5 \times 50 = 5 \times a_2$$

$$a_4 = 5 \times \dots = 5 \times a_3$$

∴ The recurrence relation for the given sequence is

$$a_n = 5a_{n-1}, n \geq 1, a_0 = 2$$

The general soln is

$$a_n = 5^n a_0 = 5^n \times 2 = 2(5)^n.$$

— x —

⑤ Find the first 5 terms of the sequence defined by $a_n = 6a_{n-1}$, $a_0 = 2$.

Sol: Given

$$a_n = 6a_{n-1}$$

Put $n=1$, $a_1 = 6a_0 = 6 \times 2 = 12$

Put $n=2$, $a_2 = 6a_1 = 6 \times 12 = 72$

Put $n=3$, $a_3 = 6a_2 = 6 \times 72 = 432$

Put $n=4$, $a_4 = 6a_3 = 6 \times 432 = 2592$

Put $n=5$, $a_5 = 6a_4 = 6 \times 2592 = 15552$

& so on.

— x —

Second-order Linear Homogeneous R.R:

Consider a method of solving R.R of the form $c_n a_n + c_{n-1} a_{n-1} + c_{n-2} a_{n-2} = 0$, $n \geq 2$
 $\hookrightarrow (1)$

where C_n, C_{n-1}, C_{n-2} are real constants with $C_n \neq 0$. A relation of this type is called a 2nd order linear homogeneous R.R with constant coefficients.

Put $a_n = Ck^n$ in (1) we get-

$$C_n Ck^n + C_{n-1} Ck^{n-1} + C_{n-2} Ck^{n-2} = 0.$$

Put $n=2$,

$$C_n k^2 + C_{n-1} k + C_{n-2} = 0. \rightarrow (2)$$

Equation (2) is called an auxiliary equation or the characteristic equation for the relation (1)

The ~~A.E~~ auxiliary roots are having 3 cases

(i) The two roots (k_1 & k_2) say of the equ (2) are real and distinct then

$$a_n = A k_1^n + B k_2^n.$$

(ii) The roots are real and equal (i.e. ^{say} $k_1 = k_2$)

$$a_n = (A + Bn) k^n.$$

(iii) The two roots are complex then

$$a_n = r^n (A \cos n\theta + B \sin n\theta)$$

where A & B are arbitrary constants.

and $r = |k_1| = |k_2| = \sqrt{p^2 + q^2}$ and

$\theta = \tan^{-1}(q/p)$

————— x —————

① Solve the R.R $a_n + a_{n-1} - 6a_{n-2} = 0$ for $n \geq 2$
Given that $a_0 = -1$ & $a_1 = 8$.

Sol:- Given

$$a_n + a_{n-1} - 6a_{n-2} = 0, \quad n \geq 2 \quad (1)$$

The characteristic equ of the R.R is

$$r^2 + r - 6 = 0.$$

$$(r+3)(r-2) = 0$$

$$\begin{array}{r} -6 \\ 3 \diamond -2 \\ 1 \end{array}$$

24 Sunday

Week 22

$r = -3, r = 2$

The General solution of the relation is

$$a_n = A K_1^n + B K_2^n.$$

$$a_n = A(-3)^n + B(2)^n \rightarrow (2)$$

Given $a_0 = -1$ & $a_1 = 8$.

To find A & B

Put $n=0$, ~~at~~ $a_0 = A(-3)^0 + B(2)^0$

$$-1 = A + B \rightarrow (3)$$

Put $n=1$,

$$a_1 = A(-3)^1 + B(2)^1$$

$$8 = -3A + 2B \rightarrow (4)$$

Solve (3) & (4)

$$(3) \times 2 \Rightarrow -2 = 2A + 2B$$

$$(4) \times 1 \Rightarrow \begin{array}{r} 8 \\ (-2) \end{array} = \begin{array}{r} -3A \\ (+) \end{array} + 2B$$

$$-10 = 5A$$

$$A = -2$$

Put $A = -2$ in (3),

$$-1 = -2 + B$$

$$-1 + 2 = B$$

$$1 = B$$

The general solution is

$$a_n = (-2)(-3)^n + (1)(2)^n.$$

_____ X _____.

② Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$

for $n \geq 2$ given that $a_0 = 5, a_1 = 12$.

Sol: Given

$$a_n - 6a_{n-1} + 9a_{n-2} = 0.$$

The auxiliary equation

$$r^2 - 6r + 9 = 0.$$

$$(r-2)(r-3) = 0$$

$$\left[\begin{array}{l} \because a_n = r^n \\ \text{Put } n=2 \end{array} \right]$$

$$\begin{array}{c} 9 \\ -3 \quad \square \quad -3 \\ \hline -6 \end{array}$$

$$r_1 = 3, r_2 = 3.$$

The roots are equal

$\therefore$ The general solution is $a_n = (A + Bn)3^n$.

Given the value $a_0 = 5, a_1 = 12$. $\hookrightarrow (1)$

Put $n=0$ in (1)

$$a_0 = (A + 0)3^0$$

$$\boxed{5 = A}$$

Put $n=1$ in (1)

$$a_1 = (A + B)3$$

$$= (5 + B)3$$

$$12 = 15 + 3B$$

$$-3 = 3B \Rightarrow \boxed{B = -1}$$

The general sol is $a_n = (5 + (-1)n)3^n$.

$$= (5 - n)3^n.$$

_____ $\times$ _____.

(3) Solve the recurrence relation $a_n = 2(a_{n-1} + a_{n-2})$

for $n \geq 2$ given that $a_0 = 1$ & $a_1 = 2$.

Sol:- The given equation

$$a_n - 2a_{n-1} + 2a_{n-2} = 0$$

$n-2$ $r-2$ $r-1$ $r^2-2r+2=0$ $\therefore a^n = r^n$
 Put $n=2$ The A.E is $r^2 - 2r + 2 = 0$ $\left[\because a^n = r^n \right]$
 $n=2$

$$\therefore K = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{2 \pm \sqrt{4-8}}{2}$$

$$= \frac{2 \pm 2i}{2} = 1 \pm i. \quad (p \pm iq)$$

The roots are complex.

The general solution is

$$a_n = r^n (A \cos n\theta + B \sin n\theta)$$

$$r = \sqrt{p^2 + q^2} = \sqrt{1+1} = \sqrt{2}$$

$$\tan \theta = \frac{p}{q} = \frac{1}{1} = 1 \Rightarrow \theta = \tan^{-1}(1) = \frac{\pi}{4}$$

$$a_n = (\sqrt{2})^n \left(A \cos n \frac{\pi}{4} + B \sin n \frac{\pi}{4} \right) \rightarrow (1)$$

Given $a_0 = 1$ & $a_1 = 2$

Put $n=0$ in (1)

$$a_0 = 1 (A \cos 0 + B \sin 0)$$

$$1 = A (1) \Rightarrow A = 1 \quad \boxed{A=1}$$

Put $n=1$ in (1)

$$a_1 = \sqrt{2} \left(A \cos \frac{\pi}{4} + B \sin \frac{\pi}{4} \right)$$

$$= \sqrt{2} \left(A \frac{1}{\sqrt{2}} + B \frac{1}{\sqrt{2}} \right)$$

$$2 = A + B$$

$$2 - 1 = B \quad \boxed{B=1}$$

The General sol is

$$a_n = (\sqrt{2})^n \left(\cos \frac{n\pi}{4} + \sin \frac{n\pi}{4} \right)$$

— α —

④ Solve the R.R $2a_{n+3} = a_{n+2} + 2a_{n+1} - a_n$ for $n \geq 0$
with $a_0 = 0, a_1 = 1, a_2 = 2$.

Soln Given

$$2a_{n+3} = a_{n+2} + 2a_{n+1} - a_n, \quad n \geq 0.$$

This eqn can be rewritten as

$$2a_n = a_{n-1} + 2a_{n-2} - a_{n-3}, \quad n \geq 3.$$

The A.E is $2a^n - a^{n-1} - 2a^{n-2} + a^{n-3} = 0.$

Put $n=3$, $2x^3 - x^2 - 2x + 1 = 0.$

$(\because a_n = x^n)$
 $n=3$

$$\begin{array}{c|cccc} 1 & 2 & -1 & -2 & 1 \\ & 0 & 2 & 1 & -1 \\ \hline & 2 & 1 & -1 & 0 \end{array}$$

$$2x^2 + x - 1 = 0.$$

[If $ax^2 + bx + c = 0$ then

Here $a=2, b=1, c=-1.$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-1 \pm \sqrt{1 - 4 \times 2 \times -1}}{2 \times 2}$$

$$= \frac{-1 \pm \sqrt{1+8}}{4} = \frac{-1 \pm 3}{4}$$

$$x_1 = \frac{-1-3}{4}, \quad x_2 = \frac{-1+3}{4} = \frac{2}{4} = \frac{1}{2}$$

$$= -1$$

$$\therefore x_1 = 1, \quad x_2 = -1, \quad x_3 = \frac{1}{2}.$$

The general solution is

$$a_n = A(1)^n + B(-1)^n + C\left(\frac{1}{2}\right)^n \rightarrow (1)$$

Given $a_0 = 0, a_1 = 1, a_2 = 2.$

Put $n=0$ in (1)

$$a_0 = A + B + C \Rightarrow A + B + C = 0 \rightarrow (2)$$

Put $n=1$ in (1)

$$a_1 = A - B + \frac{1}{2}C \Rightarrow A - B + \frac{1}{2}C = 1 \rightarrow (3)$$

Put $n=2$ in (1)

$$a_2 = A + B + \frac{1}{4}C \Rightarrow A + B + \frac{1}{4}C = 2 \rightarrow (4)$$

from (2) & (3) $2A + \frac{3}{2}C = 1 \rightarrow (5)$

from (3) & (4) $2A + \frac{3}{4}C = 3 \rightarrow (6)$

$$\begin{array}{r} (-) \quad (-) \quad (-) \\ \hline \end{array}$$

$$\frac{3}{4}C = -2$$

$$\boxed{C = -\frac{8}{3}}$$

Subst it in (5)

$$2A + \frac{3}{2}\left(-\frac{8}{3}\right) = 1$$

$$2A - 4 = 1 \Rightarrow 2A = 5 \Rightarrow \boxed{A = \frac{5}{2}}$$

Subst it A & C in (2)

$$\frac{5}{2} + B - \frac{8}{3} = 0$$

$$\boxed{B = \frac{1}{6}}$$

The g.s is $a_n = \left(\frac{5}{2}\right)(1)^n + \left(\frac{1}{6}\right)(-1)^n - \frac{8}{3}\left(\frac{1}{2}\right)^n$

⑤ Find the G.S of the R.R $a_n + a_{n-3} = 0$, $n \geq 3$.

Sol:- Given $a_n + a_{n-3} = 0$.

A.E is $x^n + x^{n-3} = 0$ $[\because a^3 + b^3 = (a+b)(a^2 - ab + b^2)]$

Put $n=3$, $x^3 + x^0 = 0$

$x^3 + 1 = 0 \Rightarrow (x+1)(x^2 - x + 1) = 0$

$x+1=0 \Rightarrow x=-1$, $x^2 - x + 1 = 0$

$x = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times 1}}{2}$

$= \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$

$\therefore$ The roots are

$K_1 = -1$, $K_2 = \frac{-1}{2} \pm i\frac{\sqrt{3}}{2}$

The G.S is $a_n = A(-1)^n + r^n \{ B \cos n\theta + C \sin n\theta \}$

$r = \sqrt{p^2 + q^2} = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$

$\theta = \tan^{-1}(q/p) = \tan^{-1}\left(\frac{\sqrt{3}/2}{-1/2}\right) = \tan^{-1}(\sqrt{3}) = \pi/3$

The general sol is

$a_n = A(-1)^n + (1)^n \left\{ B \cos \frac{n\pi}{3} + C \sin \frac{n\pi}{3} \right\}$

⑥ Solve the R.R $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$

Given $a_0 = 1$, $a_1 = 4$, $a_2 = 28$. //

Solutions of Inhomogeneous Recurrence Relations of 2nd and higher order.

The equation of the form

$$C_n a_n + C_{n-1} a_{n-1} + C_{n-2} a_{n-2} + \dots + C_{n-k} a_{n-k} = f(n) \quad \rightarrow (1)$$

for $n \geq k \geq 2$ where $C_n, C_{n-1}, \dots, C_{n-k}$ are real constants with $C_n \neq 0$ and $f(n)$ is a given real-valued function of n .

A General solution of the R.R (1) is given by

$$a_n = a_n^{(h)} + a_n^{(p)} \rightarrow (2)$$

Particular solution

① Suppose $f(n)$ is a polynomial of degree q and 1 is not a root of the characteristic equation of the homogeneous part of the relation (1)

$$a_n^{(p)} = A_0 + A_1 n + A_2 n^2 + \dots + A_q n^q$$

where $A_0, A_1, \dots, A_q$ are constants using the fact that $a_n = a_n^{(p)}$ satisfies the relation (1)

② Suppose $f(n)$ is a polynomial of degree q and 1 is a root of multiplicity m of the char. eqn. of the homogeneous part of the relation (1)

$$a_n^{(P)} = n^m \{ A_0 + A_1 n + A_2 n^2 + \dots + A_r n^r \}$$

③ Suppose $f(n) = \alpha b^n$ where α is a constant and b is not a root of the char equ of the homogeneous part of the relation (1). Then $a_n^{(P)}$ is taken in the form

$$a_n^{(P)} = A_0 b^n$$

④ Suppose $f(n) = \alpha b^n$ where α is a constant and b is a root of multiplicity m of the char equ of the homogeneous part of the relation (1). Then $a_n^{(P)}$ is taken in the form

$$a_n^{(P)} = A_0 n^m b^n$$

_____ α _____

① Solve the recurrence relation

$$a_n + 4a_{n-1} + 4a_{n-2} = 8 \quad \text{for } n \geq 2.$$

Sol:-

the char equ

$$a^k + 4a^{k-1} + 4a^{k-2} = 0$$

Put $k=2$, $a^2 + 4a + 4 = 0$

$$(a+2)(a+2) = 0$$

$$a_1 = -2, \quad a_2 = -2$$

$$\begin{array}{c} 4 \\ 2 \swarrow \searrow 2 \\ 4 \end{array}$$

The roots are equal

$$a_n^{(h)} = (A + Bn)(-2)^n \rightarrow (1)$$

To find $a_n^{(p)}$:

Roots of the given relation (constant)

$$a_n^{(p)} = A_0 \text{ (or) } a_0$$

Put $a_n^{(p)}$ in the given eqn

$$A_0 + 4A_0 + 4A_0 = 8$$

$$9A_0 = 8$$

$$A_0 = \frac{8}{9}$$

$$\therefore a_n^{(p)} = \frac{8}{9} \rightarrow (2)$$

The General solution is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$= (A + Bn)(-2)^n + \frac{8}{9} \rightarrow (3)$$

Given

$$a_0 = 1, \quad a_1 = 2$$

$$\text{put } n=0, \quad a_0 = (A + 0)(-2)^0 + \frac{8}{9}$$

$$1 = A + \frac{8}{9}$$

$$1 - \frac{8}{9} = A \Rightarrow \boxed{A = \frac{1}{9}}$$

$$\text{put } n=1, \quad a_1 = (A + B)(-2)^1 + \frac{8}{9}$$

$$2 = \left(\frac{1}{9} + B\right)(-2) + \frac{8}{9}$$

$$\boxed{B = -\frac{2}{3}}$$

The General solution is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$= \left(\frac{1}{9} - \frac{2}{3}n \right) (-2)^n + \frac{8}{9}$$

which is the required solution.

_____ x _____.

(2) Solve the recurrence relation

$$a_{n+2} + 3a_{n+1} + 2a_n = 3^n \text{ for } n \geq 0$$

given $a_0 = 0, a_1 = 1$.

Sol:-

[Page: 30]

③ condition.

The char eqn is $a^2 + 3a + 2 = 0$

$$(a+2)(a+1) = 0$$

$$a_1 = -2, a_2 = -1.$$

$$\begin{array}{c} 2 \\ \swarrow \downarrow \searrow \\ 2 \quad 1 \quad 3 \end{array}$$

$$\therefore a_n^{(h)} = A(-2)^n + B(-1)^n$$

The R.H.S of given eqn is 3^n

$$a_n^{(p)} = A_0 3^n = (A_0 b^n)$$

Put a_n in the given eqn we get

$$A_0 3^{n+2} + 3A_0 3^{n+1} + 2A_0 3^n = 3^n$$

$$3^n [A_0 9 + 3A_0 3 + 2A_0] = 3^n$$

$$3^n [20A_0] = 3^n$$

$$20A_0 = 1$$

$$\boxed{A_0 = \frac{1}{20}}$$

$$\therefore a_n^{(P)} = \frac{1}{20} (3)^n$$

The general soln is $a_n = A(-2)^n + B(-1)^n + \frac{1}{20}(3)^n$

Given $a_0 = 0, a_1 = 1 \rightarrow (1)$

put $n=0, a_0 = A + B + \frac{1}{20}$

$$0 = A + B + \frac{1}{20}$$

$$A + B = -\frac{1}{20} \rightarrow (2)$$

put $n=1, a_1 = -2A - B + \frac{3}{20}$

$$1 - \frac{3}{20} = -2A - B$$

$$\frac{17}{20} = -2A - B \rightarrow (3)$$

Solve (2) & (3), we get

$$A + B = -\frac{1}{20}$$

$$-2A - B = \frac{17}{20}$$

$$\underline{-A = \frac{16}{20}}$$

$$\boxed{A = -\frac{4}{5}}$$

put $A = -\frac{4}{5}$ in (2)

$$-\frac{4}{5} + B = -\frac{1}{20}$$

$$B = -\frac{1}{20} + \frac{4}{5} = \frac{-1+16}{20} = \frac{15}{20} = \frac{3}{4}$$

$$B = \frac{3}{4}$$

$\therefore$ The required soln is

$$a_n = \frac{3}{4}(-1)^n + \left(-\frac{4}{5}\right)(-2)^n + \frac{1}{20}3^n$$

③ Solve the recurrence relation

$$S(n+2) - 10S(n+1) + 21S(n) = 3n^2 - 2$$

Sol:- The A.E is

$$a^{n+2} - 10a^{n+1} + 21 = 0$$

17 Sunday Put $n=0$, $a^2 - 10a + 21 = 0$

$$(a-3)(a-7) = 0$$

$$a = 3, a = 7$$

$$\therefore a_n^{(h)} = A(3)^n + B(7)^n \rightarrow (1)$$

To find $a_n^{(p)}$

Given RHS of the relation is a Polynomial of degree 2. $a_n^{(p)} = d_0 + d_1n + d_2n^2 \rightarrow (2)$

$$\text{Take } S(k) = d_0 + d_1k + d_2k^2$$

$$S(k+1) = d_0 + d_1(k+1) + d_2(k+1)^2$$

$$S(k+2) = d_0 + d_1(k+2) + d_2(k+2)^2$$

$$\begin{aligned} & [d_0 + d_1(k+2) + d_2(k+2)^2] - 10[d_0 + d_1(k+1) \\ & \quad + d_2(k+1)^2] \\ & \quad + 21[d_0 + d_1k + d_2k^2] = 3n^2 - 2 \end{aligned}$$

$$\begin{aligned} & (d_0 + d_1k + 2d_1 + k^2d_2 + 4d_2 + 4kd_2) \\ & - 10(d_0 + d_1k + d_1 + k^2d_2 + 2kd_2 + d_2) \\ & + 21d_0 + 21d_1k + 21d_2k^2 = 3n^2 - 2. \end{aligned}$$

$$\begin{aligned} & d_0 + d_1k + 2d_1 + k^2d_2 + 4d_2 + 4kd_2 - 10d_0 - 10d_1k \\ & - 10d_1 - 10k^2d_2 - 20kd_2 - 10d_2 + 21d_0 \\ & + 21d_1k + 21d_2k^2 = 3k^2 - 2. \end{aligned}$$

$$\begin{aligned} & (d_2 - 10d_2 + 21d_2)k^2 + (d_1 + 4d_2 - 10d_1 - 20d_2 \\ & \quad + 21d_1)k \end{aligned}$$

$$\begin{aligned} & + (d_0 + 2d_1 + 4d_2 - 10d_0 - 10d_1 - 10d_2 + 21d_0) \\ & = 3k^2 - 2. \end{aligned}$$

$$\begin{aligned} & 12d_2n^2 + (12d_1 - 16d_2)n + (12d_0 - 8d_1 - 6d_2) \\ & = 3n^2 - 2. \end{aligned}$$

Equating the coeff of n^2

$$12d_2 = 3 \Rightarrow \boxed{d_2 = \frac{1}{4}}$$

Equating the coeff of n

$$12d_1 - 16d_2 = 0$$

$$12d_1 = 16\left(\frac{1}{4}\right) = 4 \Rightarrow \boxed{d_1 = \frac{1}{3}}$$

Week 23

Equating the coefficient of constant (36)

$$12d_0 - 8d_1 - 6d_2 = -2$$

$$12d_0 - 8\left(\frac{1}{3}\right) - 6\left(\frac{1}{4}\right) = -2$$

$$12d_0 - \frac{8}{3} - \frac{3}{2} = -2$$

$$12d_0 = -2 + \frac{8}{3} + \frac{3}{2}$$

$$= \frac{-12 + 16 + 9}{6} = \frac{-12 + 25}{6} = \frac{13}{6}$$

$$12d_0 = \frac{13}{6}$$

$$d_0 = \frac{13}{72}$$

substn in ② we get

$$a_n^{(P)} = \frac{13}{72} + \frac{1}{3}n + \frac{1}{4}n^2$$

7 Sunday

Week 24

The general soln is

$$a_n = a_n^{(h)} + a_n^{(P)}$$

$$= A(3)^n + B(7)^n + \left(\frac{13}{72} + \frac{n}{3} + \frac{n^2}{4}\right)$$

④ Solve the recurrence relation

$$a_{n+2} - 4a_{n+1} + 3a_n = -200, \quad n \geq 0 \text{ and}$$

$$a_0 = 3000, \quad a_1 = 3300.$$

Sol:- Given

$$a_{n+2} - 4a_{n+1} + 3a_n = -200 \quad (1) \quad [a^0 = 1]$$

The char eqn

$$a^{n+2} - 4a^{n+1} + 3a^n = 0$$

Put $n=0$ $a^2 - 4a + 3 = 0$

$$a = 1, 3.$$

$$\therefore a_n^{(h)} = A(1)^n + B(3)^n \rightarrow (1)$$

The RHS of the given R.R is a constant

~~then replace a_n, a_{n+1}, a_{n+2} by d_0~~

$$\cancel{d_0} - \cancel{d_0} + \cancel{d_0} = \cancel{-200}$$

Since 1 is a (simple) root of the char eqn

and the RHS of the given relation is a polynomial of degree 0, we seek $a_n^{(p)}$ in the form

$$a_{n+1}^{(p)} = A_0(n+1)$$

$$a_{n+2}^{(p)} = A_0(n+2)$$

$$a_n^{(p)} = A_0 n. \rightarrow (2)$$

$$\left[\begin{array}{l} \text{Prob-1} \\ a_{n+2} = d_0 \\ a_{n+1} = d_0 \\ a_n = d_0. \end{array} \right]$$

Given $a_{n+2} - 4a_{n+1} + 3a_n = -200$

$$A_0(n+2) - 4A_0(n+1) + 3A_0 n = -200$$

$$A_0 n + 2A_0 - 4A_0 n - 4A_0 + 3A_0 n = -200$$

$$-2A_0 = -200$$

$$A_0 = 100$$

Substituting A_0 in (2) we get

$$a_n^{(p)} = 100n.$$

$\therefore$ The G.S for a_n is

$$a_n = A + B(3)^n + 100n \rightarrow (3)$$

Given $a_0 = 3000$, $a_1 = 3300$

put $n=0$ in (3)

$$a_0 = A + B(3)^0 + 100(0)$$

$$3000 = A + B \rightarrow (4)$$

put $n=1$ in (3)

$$a_1 = A + 3B + 100$$

$$3300 = A + 3B + 100 \rightarrow (5)$$

Solve (4) & (5)

$$A = 2900, B = 100$$

The required soln is

$$a_n = 2900 + 100(3)^n + 100n.$$

working Rules $\times$

① If the RHS is $a_0 + a_1 k$ replace $T(k)$ by

$a_0 + a_1 k$, $T(k-1)$ by $a_0 + a_1(k-1)$

$T(k-2)$ by $a_0 + a_1(k-2)$ etc.

② when the RHS is ca^k and a coincides with a char root the above method fails. when a is a simple root of the char equ take

do $K a^K$ when a is a double root of the char equ take $do K^2 a^K$.

———— α ————

① solve $S(K) - 7S(K-1) + 10S(K-2) = 6 + 8K$
with $S(0) = 1$ & $S(1) = 2$.

② Solve $S(K) - 3S(K-1) - 4S(K-2) = 4^K, K \geq 2$
using Recurrence relation.

③ solve $S(K) - 4S(K-1) + 4S(K-2) = 3K + 2^K,$
 $K \geq 2$ with $S(0) = 1, S(1) = 1$

———— α ————